



Agentic AI-Based Dynamic Tariff Optimization

A six-phase pipeline transforming raw EV charging data into intelligent, demand-responsive pricing — from ingestion to final business metrics.

OPEN PROJECT SOCIETY OF BUSINESS IITR 2026

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PHASE 1

Data Ingestion & Preprocessing

Two distinct datasets unified into a clean, analytically ready base.

ACN-Data • US Workplace Sessions

- Load ~30,000 sessions from JSON/Excel
- Parse timestamps; calculate session, charging, and idle hours
- Drop invalid rows; add temporal and economic features

UrbanEV Data • Shenzhen Urban Network

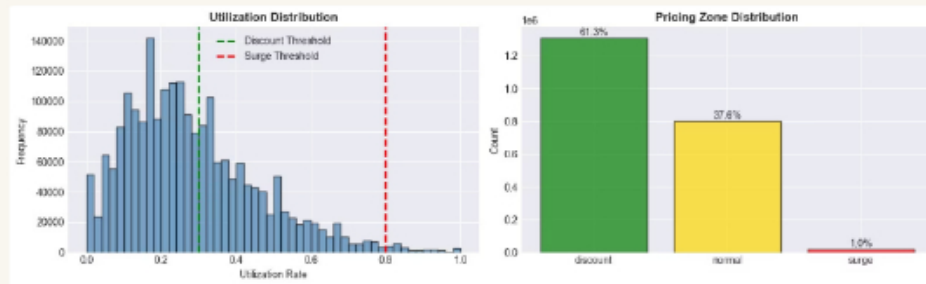
- Ingest and melt CSVs (time, occupancy, volume, price, info) to long format
- Calculate **Charger Utilization Rate** (Occupancy / Total Chargers)
- Discretize into pricing zones: Discount ($\leq 30\%$), Normal (30–80%), Surge ($\geq 80\%$)

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PHASE 2

Exploratory Data Analysis

Baseline behavioral profiling justifies the need for dynamic pricing and identifies high-impact intervention windows.



Utilization Distribution & Pricing Zone Distribution

Behavioral Profiling

ACN shows sharp morning commuter peaks (workplace behavior). UrbanEV exhibits sustained daytime usage (urban behavior).

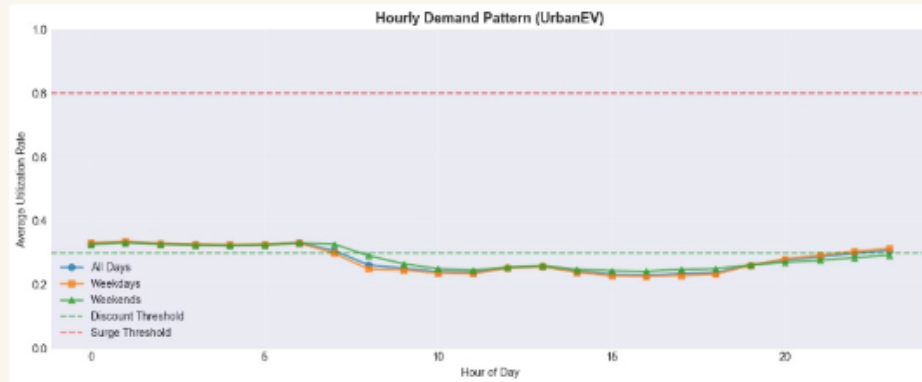


ACN Sessions by Hour & Site and by Day of Week

Volatility Analysis

Grouping utilization into **Peak**, **Shoulder**, and **Off-Peak** periods reveals where pricing interventions yield the highest revenue impact.

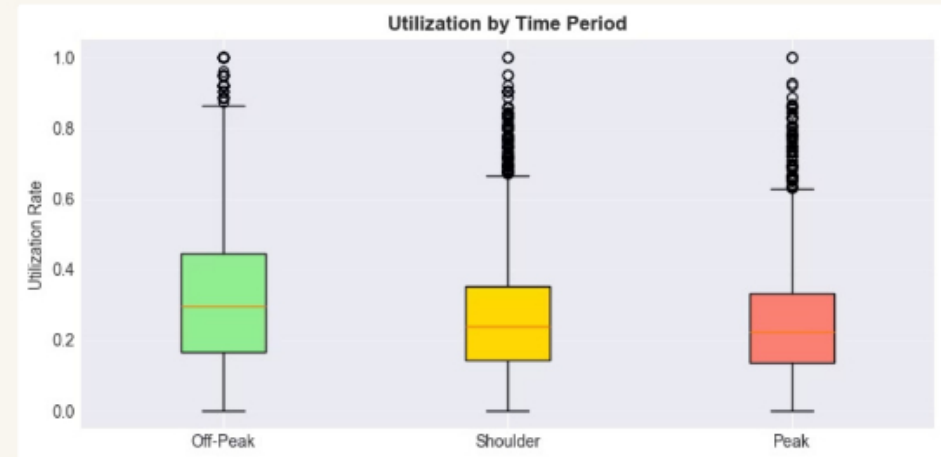
EDA Visualizations (Continued)



Hourly Demand Pattern (UrbanEV)

Demand Pattern

UrbanEV demand shows distinct hourly fluctuations, highlighting the periods where pricing and capacity planning matter most.



Utilization by Time Period

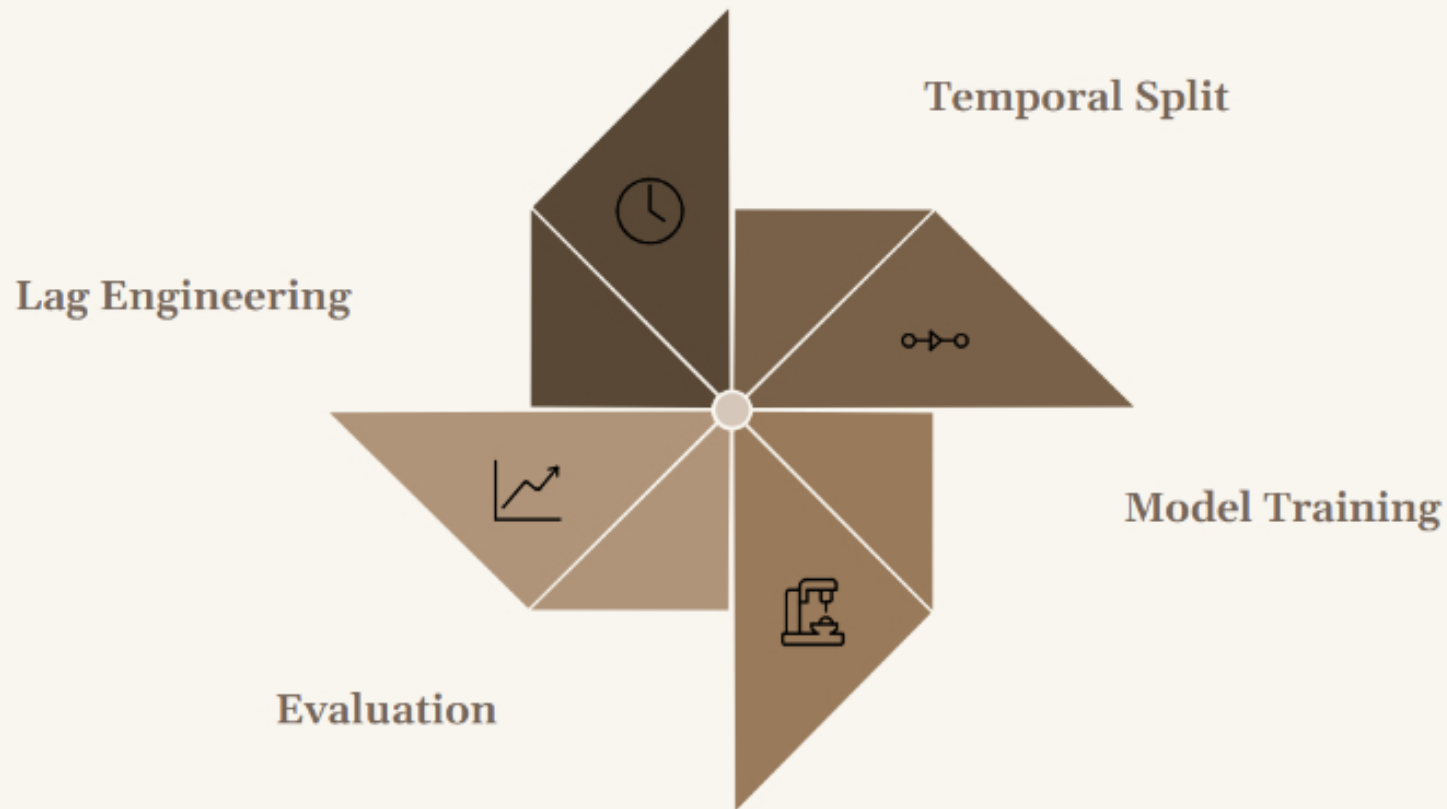
Time Period Analysis

Grouping utilization into peak, shoulder, and off-peak periods reveals where interventions can improve revenue and balance demand.

PHASE 3

The Demand Prediction Agent

Best Model Performance



Random Forest

RMSE-0140

MAE - 0.0052

R^2 -0.9937

The Dynamic Tariff Pricing Agent

Translate demand forecasts into automated pricing signals and quantify revenue impact against a static Rs 15/kWh baseline.

Pricing Logic

Surge $\geq$
80%

Rs 22.50/kWh

Normal
30–80%

Linear scale

Discount $\leq$
30%

Rs 12.75/kWh

Localized Mapping & Revenue

- UrbanEV: Apply dynamic tariffs to test set; measure **Off-Peak Uplift**
- ACN: Build a localized utilization proxy from its own historical volume — no cross-market curve transfer
- Compare total dynamic revenue vs. static baseline → **Revenue Gain %**

Final Tariff Results

Base tariff

Rs 15.00/kWh

Revenue Gain
% (ACN)

7.64%

Charger
Utilization
Rate
(UrbanEV)

0.2908

Off-Peak
Uplift
(UrbanEV)

-0.03%

Pricing
Efficiency
(ACN)

Rs 16.15/kWh
(7.64%
improvement)

Monitoring, Learning & Export

Phase 5 · Monitoring & Learning Agent

Simulate live outcomes with a **PED coefficient of -0.30** (10% price increase $\rightarrow$ 3% demand drop).

- **Avg Waiting Time Reduction (UrbanEV Simulated):** 94.41%
- **Avg Queue Proxy:** 0.0166 baseline $\rightarrow$ 0.0009 simulated
- **Episodes tracked (UrbanEV Test Set):** 144
- **Pricing Efficiency:** revenue per kWh vs. static baseline
- **Episode Dashboard:** hourly actual vs. predicted utilization, prediction error, and revenue gains

Tariff Zone	Baseline Avg kWh	Simulated Avg kWh
discount	8.69	9.08
normal	9.23	8.75
surge	9.10	7.74

Phase 6 · Export & Reporting

Package final deliverables to **outputs/**:

- demand_predictions.csv
- pricing_output.csv
- monitoring_metrics.csv
- summary_report.csv